

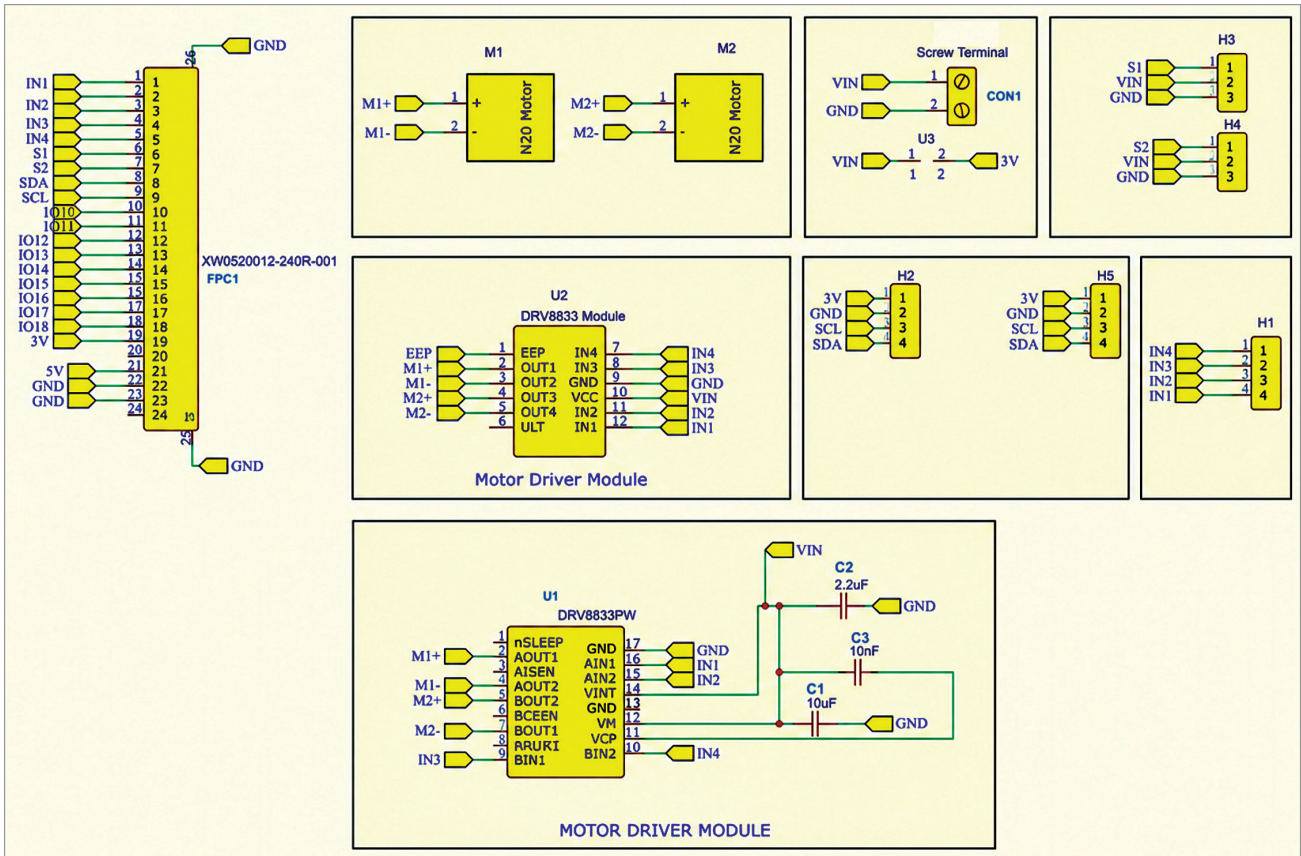


Fig. 2: Circuit diagram

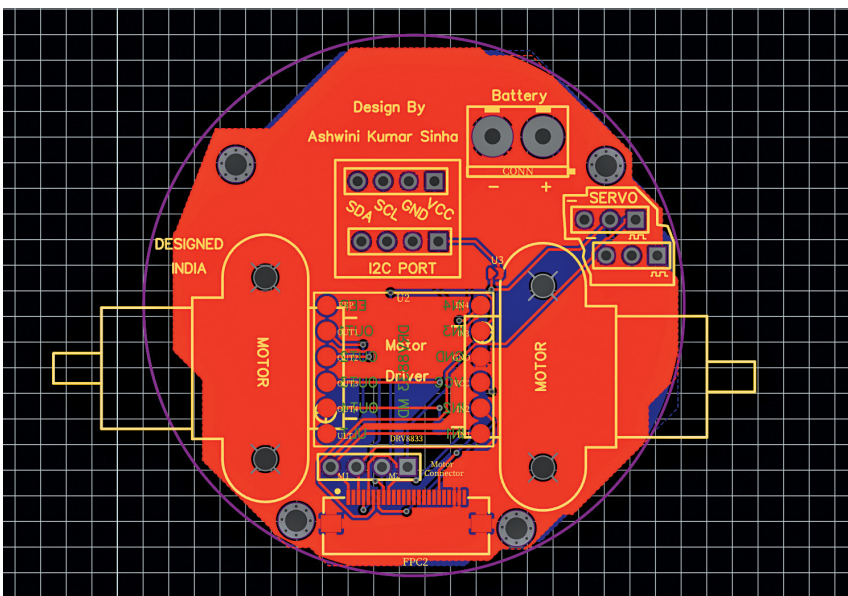


Fig. 3: PCB design of the motor-driver robot chassis

FPC cable. For other microcontrollers, only four motor-control signals are required, and a custom FPC adaptor can be implemented if needed.

The system is based on the

DRV8833 dual motor driver IC and includes two servo connectors and two I²C expansion connectors. These interfaces allow direct connection of displays and sensors without

additional wiring. A dedicated battery input is provided, and configurable solder jumpers allow users to either share a common power source between the motor driver and microcontroller or use separate supplies for greater flexibility.

In addition to simplifying control and connectivity, the PCB also serves as the robot's structural chassis. It incorporates integrated N20 motor-mounting slots and four expansion-mounting holes, enabling easy stacking of additional layers and modules such as lidar, radar, cameras, displays, and other sensors. This modular architecture allows the robot to be customised for a wide range of applications while maintaining a compact footprint.

Fig. 1 shows the author's prototype and the developed PCB. The components required to build the integrated motor-driver robot chassis are listed in Bill of Materials table.